

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination : Semester Final
 Duration: **2 Hours**

Semester: **Fall 2025**
 Full Marks: **70**

CSE421 / EEE465 : Computer Networks

Answer **Sections A, B and C** as per instructions given. (**Pages: 4**)

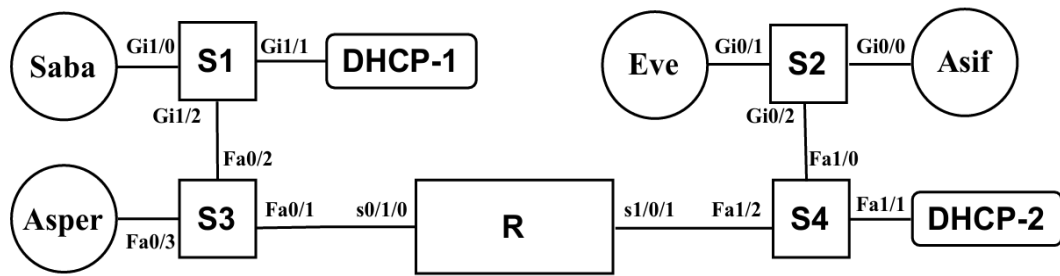
Figures in the right margin indicate marks.

Name:	ID:	Section:
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SECTION A [All questions of this section are MANDATORY] - 40 MARKS

Q1 [CO3]	A network consists of two subnets, referred to as Network A and Network B. Network A and Network B are consecutive subnets within the same major address block (i.e., there are no other subnets between them). Network A supports 1022 hosts. A PC in Network A is assigned the first usable IP address of its subnet. This PC sends an IP packet with the following addresses: Source IP address: 172.16.8.1 Destination IP address: 172.16.13.255. The packet is received by all devices in Network B. I. Determine the network (subnet) address of Network B. II. Determine the maximum number of hosts that Network B can support. III. Given the IP block 10.192.128.0/18. Apply VLSM to identify the sub-network addresses for each subnet present in the figure shown. The number of hosts given in the topology only includes the end devices.	1048 hosts 250 hosts 62 hosts R1 R2 R3 SW2	4 + 2 + 10
Q2 [CO3] [CO3] [CO3] [CO2]	A network-layer PDU of 4680 bytes, including a 20-byte header, is fragmented for transmission over a link with an MTU of X bytes. The difference between the fragment offset values of two consecutive fragments is 75. The fragment offset of the first fragment is 0. I. Calculate the value of MTU (X). II. Calculate the fragment offset of the 4th fragment. III. Calculate the data size of the last fragment. IV. If the DF value of the original datagram was set, how was the packet sent to its destination?		3 + 3 + 4 + 3

Q3
[CO2]



Saba and Eve have just connected to their networks. All switches have empty MAC tables; MAC-entry aging time is 2 minutes. Consider the MAC address of each host to be its name (e.g., ASIF).

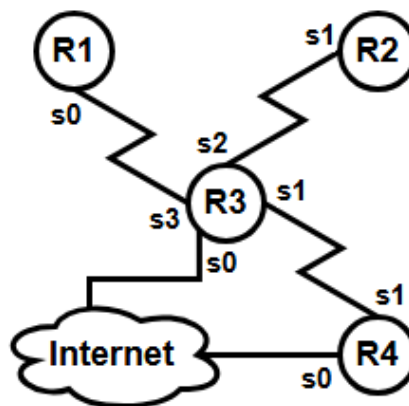
- I. Saba's PC sends its first DHCP message to obtain an IP address. What source IP address is used in this packet?
- II. Asper connects at the same time as Saba and sends a DHCP request. How can DHCP-1 distinguish Saba's request from Asper's request even though both requests are broadcast?
- III. Eve has used an address from DHCP-2 for 5 hours. Her renewal attempt to DHCP-2 fails, but she then successfully obtains a new address from DHCP-1. Briefly explain how Eve is able to obtain an IP address from DHCP-1.
- IV. Asper intends to send a packet to Eve. Since Eve is on a different subnet, Asper must send the packet to Router R. Name the specific link-layer requests and responses that are exchanged between Asper and Router R to make this possible?

END OF SECTION A

[CO3] **SECTION B** [Answer ANY TWO out of THREE in this section] - 12 MARKS

Q4 Given a topology below. Each router has a LAN connected to its f0 interface.

- I. **Configure** a non-recursive static route to R2 LAN from R4.
- II. Assume R4 is routing to all unknown routes to the internet via s0. **Configure** a floating static route for this route.



Device	Int.	IP	Network
R1	f0	.1	10.10.10.0/24
	s0	.1	10.14.10.0/30
R2	f0	.1	10.10.11.0/24
	s1	.1	10.15.10.0/30
R3	f0	.1	10.10.12.0/24
	s0	.2	10.17.10.0/30
	s1	.2	10.16.10.0/30
	s2	.2	10.15.10.0/30
	s3	.2	10.14.10.0/30
R4	f0	.1	10.10.13.0/24
	s0	.1	10.18.10.0/30
	s1	.1	10.16.10.0/30

Q5 I. **Expand** the following IPv6 Addresses in their Hexadecimal form.

3
+

3
+

	A. FE80::1A2B B. 2001:DB8:1234::10 C. ::25 II. A PC sends a packet to 224.10.10.5, but only some PCs receive it. Explain why.	3
Q6	Using the given diagram in Q3, answer the following questions. I. Asif wants to send an IP packet to Eve, but does not yet know Eve's MAC address. What frame does Asif generate (source and destination MAC), and how do the switches S1, S2, S3, and S4 process this frame? II. After Eve replies to Asif, show the MAC address tables of all the switches.	3 + 3
END OF SECTION B [CO2] SECTION C [Answer <u>ANY THREE</u> out of <u>FIVE</u> in this section] - 18 MARKS		
Q7	I. Apart from limiting the number of hops, explain how the Time to Live (TTL) field helps prevent network problems. What would happen to the Internet if the TTL field did not exist? II. An attacker discovers a network with 200 active devices. He sends one ICMP Echo Request in such a way that the target victim receives 200 ICMP Echo Replies. How is it possible?	3 + 3
Q8	Suppose Router R1, R3 in Q4's topology are using the Distance Vector Routing Algorithm, while routers R2 and R4 are using the Link State routing Protocol. Suddenly, the link between R1 and R3 fails. I. Explain whether this link failure will cause the count-to-infinity problem. II. Once the link between R1 and R3 fails, all routers have only their directly connected networks in their routing tables. Explain the reason.	3 + 3
Q9	Assume Host A & D inside a LAN trying to communicate with a remote host B through an edge router that performs PAT. The edge router has 220.20.20.17/30 in its PAT pool. A packet from host A to host B leaves with source IP 192.168.2.105, source port 52157, destination IP 6.6.6.6, and destination port 344. A packet from host D to host B leaves with source IP 192.168.2.106, source port 52157, destination IP 6.6.6.6, and destination port 344. I. Identify the source IP and port address of the packets sent by host A and host D when they leave the edge router for host B. II. Suppose a remote host C on the Internet wants to initiate new communication with host A. Explain why or why not this would be possible with the current setup. If not, how can it be made possible?	3 + 3
Q10	Scenario 1: An organization needs IPv6 devices to communicate with IPv4 devices during the transition period. Scenario 2: A service is deployed using the same IPv6 address on multiple routers. Clients always reach the nearest router automatically.	1 + 2 + 3

	I. Name the IPv6 transition technique used in Scenario 1. II. What IPv6 addressing mechanism is used in Scenario 2? Is there any difference between this mechanism and multicast addressing? III. IPv6 does not have an Options field like IPv4. Since the IPv6 base header is fixed at 40 bytes, mention how IPv6 carries additional information when required.	
Q11	Rahim is learning about link layer communication. He notices that sometimes a frame sent using a MAC address reaches only one device, while at other times it reaches a group of devices. I. Rahim wants to understand how to determine whether a given MAC address is unicast or multicast. Explain how he can identify this from the MAC address. II. His friend remarked that MAC addresses are hierarchical and portable. Justify whether his remark about MAC addresses is correct.	3 + 3

END OF SECTION C

===== **THE END** =====

Why don't packets ever get lost at home?
They always know their default gateway.